

Arithmetic Sequences as Linear Functions

Answer Key

Algebra 1 Sequences and Linear Modeling

Quick Answers

1. 33	4. 11	7. 103	10. 4
2. 31	5. -13	8. 11	11. 615
3. 100	6. 348	9. 820	12. 405, 5250

Common wrong answers

1. If they got 38: used n instead of $n-1$, adding one common difference too many
 5. If they got 53: added the common difference instead of subtracting it
 6. If they got 216: multiplied the first row by the number of rows and ignored the increase
 9. If they got 1580: multiplied the last term by the number of terms instead of using the average of first and last
 11. If they got 930: used 15 rows of the last row's 62 seats instead of averaging first and last
 12. If they got 8100: multiplied the week-20 output by 20 instead of summing the twenty different weekly outputs
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1. 7th term of the sequence with $a_1 = 3$, $d = 5$.

Step 1. The question asks for one specific term, so the explicit formula $a_n = a_1 + (n - 1)d$ is the right tool; identify $a_1 = 3$, $d = 5$, $n = 7$ before substituting.

Step 2. $a_7 = 3 + (7 - 1)(5) = 3 + 30 = 33$. 33

Why $n - 1$: getting from term 1 to term 7 takes six steps, not seven. Using n instead gives 38, one common difference too many.

2. $f(10)$ for $f(n) = 3n + 1$.

Step 1. The sequence has already been packaged as a linear function of the term number, so the work is a single substitution rather than a recursion.

Step 2. $f(10) = 3(10) + 1 = 30 + 1 = 31$. 31

Reading the form: the slope 3 is the common difference, and the constant 1 is the value the line would take at $n = 0$ — one step *before* the first term, which is why it is not the first term itself.

3. $1 + 3 + 5 + \cdots + 19$.

Step 1. This is a total, not a term, so pair the ends: every pair of terms equally far from the two ends has the same sum, which turns the addition into a multiplication.

Step 2. $1 + 19 = 20$, $3 + 17 = 20$, $5 + 15 = 20$, and so on. With 10 terms there are 5 such pairs, each worth 20.

Step 3. $5 \times 20 = 100$. 100

General form: $S_n = \frac{n(a_1 + a_n)}{2} = \frac{10(1 + 19)}{2} = 100$ — the same pairing written once and for all.

4. Which term of 5, 9, 13, ... is 45?

Step 1. Here the term *value* is known and the term *number* is not, so write the explicit formula as an equation and solve for n rather than listing terms.

Step 2. $5 + (n - 1)(4) = 45$, so $4n + 1 = 45$, then $4n = 44$ and $n = 11$. 11

Sanity check: n must be a positive whole number. If it came out as 10.5, the value 45 would simply not be in the sequence.

5. 12th term of a sequence starting at 20 and decreasing by 3.

Step 1. “Decreases by 3” makes the common difference negative, so record $d = -3$ before substituting — the formula itself does not change.

Step 2. $a_{12} = 20 + (12 - 1)(-3) = 20 - 33 = -13$. -13

The common error: using $d = +3$ gives 53, and a sequence that was described as decreasing would have grown. Check the sign against the story.

6. Auditorium: 12 rows, 18 seats in row 1, 2 more per row.

Step 1. Part (b) asks for a total across all rows, so this is a sum; but the sum formula needs the last term, so find that first.

Step 2. Last row: $a_{12} = 18 + (12 - 1)(2) = 18 + 22 = 40$ seats.

Step 3. Total: $S_{12} = \frac{12(18 + 40)}{2} = 6 \times 58 = 348$. 348

The trap: $12 \times 18 = 216$ counts every row as the front row and throws away the growth entirely.

7. 7, 11, 15, ... as $f(n) = 4n + 3$; find $f(25)$.

Step 1. Identify the two constants from the sequence itself: the terms rise by 4 each step, so the slope is 4, and the constant is chosen so that $f(1)$ returns the first term.

Step 2. $f(1) = 4(1) + 3 = 7$, which confirms the form. Then $f(25) = 4(25) + 3 = 100 + 3 = 103$. 103

Where the numbers come from: the 4 is the common difference (the slope), and the 3 is $a_1 - d = 7 - 4$, the imaginary “term zero”.

8. Priya’s savings reach 500 dollars in which month?

Step 1. The balance is arithmetic with $a_1 = 150$ and $d = 35$, and the unknown is the month number, so solve the explicit formula for n .

Step 2. $150 + (n - 1)(35) = 500$, so $35n + 115 = 500$, then $35n = 385$ and $n = 11$. 11

Check: $n = 11$ is a whole number, so the balance hits 500 exactly at the end of month 11. Had n come out fractional, the answer would be the next whole month, since deposits arrive in whole steps.

9. Sum of the first 20 terms of 3, 7, 11, ...

Step 1. The sum formula needs the last term, so compute a_{20} before summing anything.

Step 2. $a_{20} = 3 + (20 - 1)(4) = 3 + 76 = 79$.

Step 3. $S_{20} = \frac{20(3 + 79)}{2} = 10 \times 82 = 820$. 820

The trap: $20 \times 79 = 1580$ treats every term as if it were the largest one. The correct factor is the *average* of the first and last terms, 41, not the last term itself.

10. 4th term 14, 9th term 34 — find d and a_1 .

Step 1. Neither the first term nor the common difference is given, but the gap between two known terms is made only of common differences, so count the steps rather than guessing a_1 .

Step 2. From term 4 to term 9 is $9 - 4 = 5$ steps, so $14 + 5d = 34$, giving $5d = 20$ and $d = 4$. 4

Part (c): now $a_4 = a_1 + 3d$, so $14 = a_1 + 12$ and the first term is $a_1 = 2$. The sequence is 2, 6, 10, 14, ..., and indeed its 9th term is $2 + 8(4) = 34$.

11. Theatre: 15 rows, 20 seats in row 1, 3 more per row.

Step 1. Part (a) asks for a rule, not a number, so write the explicit formula with the row number as the variable: seats in row k is $20 + (k - 1)(3)$, which simplifies to $3k + 17$.

Step 2. Last row: $20 + (15 - 1)(3) = 20 + 42 = 62$ seats.

Step 3. Capacity: $S_{15} = \frac{15(20 + 62)}{2} = \frac{15 \times 82}{2} = 615$. 615

The trap: $15 \times 62 = 930$ fills every row to the size of the back row. The average row holds 41 seats, not 62.

12. Challenge: factory output, 120 units in week 1, +15 per week.

Step 1. Part (a) asks for a single week's output and part (b) for a running total, so separate the two tools before computing: explicit formula first, sum formula second.

Step 2. Week 20: $a_{20} = 120 + (20 - 1)(15) = 120 + 285 = 405$ units. 405

Step 3. Total over 20 weeks: $S_{20} = \frac{20(120 + 405)}{2} = 10 \times 525 = 5250$ units. 5250

Part (c): multiplying the week-20 output by 20 gives 8100, which assumes every week matched the best week. The correct idea is to multiply by the *average* weekly output, $\frac{120 + 405}{2}$, which is what the sum formula does.